

ReadMe Document

This ReadMe document will help you with a quick navigation through the contents of the Pre-Reading material, Preview videos, and Individual Labs for the upcoming session.

Outcomes

After completion of this week's sessions, & activity participants will be able to :

1. Understand the concepts of Dimensionality Reduction for data visualization viz. PCA, ISOMAP ,LLE, t-SNE, etc.
2. Understand and implement the concepts of K-Means and H-Clustering.

Lecture Topics

1. Data Visualization
2. Unsupervised Learning-Clustering

Preview Videos

The Preview Videos Link for the respective sessions are available a week in advance (i.e. every Monday), to help you understand the concepts that will be covered in the upcoming lectures. There could be one or more videos depending on the lecture details. The video/videos should be viewed before attending the lecture for better understanding.

There are 7 preview videos for this session and they should be viewed as given below:

1. Data Visualization: This is an 8-minute video. From this video you will learn about data visualization and how it works. In machine learning, we use techniques like PCA, t-SNE, ISOMAP, LLE to visualize the data by transforming it into a lower dimension.
2. Unsupervised Learning (K-Means): This is a 11-minute video. From this video, you will understand about k-means. K-means is an example of unsupervised learning as the data are unlabelled. It partitions the data into k clusters, such that each cluster has samples closest to the mean of that cluster
3. Hierarchical Clustering: This is a 5-minute video. From this video, you will understand about hierarchical clustering techniques such as agglomerative hierarchical clustering (grouping data points into larger and larger clusters - eg: single Link Algorithm) and divisive hierarchical clustering (assigning all the data points to a single cluster and then partitioning the cluster into smaller and smaller clusters)

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Pre Reading Material

The pre-reading material is given on t-SNE, Isomap, K-Means and Hierarchical Clustering. You should read the material before you attend the lecture. This pre-reading will provide the gist of the lecture and facilitate your understanding during the lecture.

Reference Material

The Reference Material is given on t-SNE. You should read the material after attending the lecture. This reference material will provide further depth into the topic.

Individual Labs/Demo

1. Clustering
2. Isomap
3. t-SNE

